Eedi: forgetting threshold comparison

Fixed theta_m=0.8 and rec=50 | Seed 2024 | Six ablation models per group

Three completed groups: theta_f=0.5, 0.6, and 0.8.

Best is bold; second best is underlined, based on unrounded values.

Main-model mean across N = 5, 10, ..., 50 (descriptive only)

Metric	theta_f=0.5	theta_f=0.6	theta_f=0.8
Adaptability (Ada)	**0.670**	0.665	0.664
Novelty (NOV)	0.981	**0.984**	0.981
Ep, response threshold = 0.4	**0.058**	-0.100	0.034
Ep, response threshold = 0.5	**-0.178**	-0.442	-0.274
Ep, response threshold = 0.6	**-0.405**	-0.605	-0.561
Ep, response threshold = 0.7	**-0.615**	-0.678	-0.651

Reading the tables

Pages 2-4 compare theta_f for the main model at each N.

Pages 5-13 show all six models within each parameter group.

Ep is a simulated mastery gain; negative values indicate a simulated decrease.

All values use one random seed; no between-seed standard deviation is available.

Source ZIPs

Eedi_results_seed2024_20261006_154354.zip

Eedi_new_two_groups_seed2024_20261006_221346.zip

2CKG4ER across forgetting thresholds

theta_m=0.8, rec=50; same seed and model; higher values rank better

(a) Adaptability (Ada)

N	theta_f=0.5	theta_f=0.6	theta_f=0.8
5	**0.672**	0.664	0.664
10	**0.671**	0.661	0.661
15	**0.669**	0.664	0.662
20	**0.670**	0.665	0.663
25	**0.671**	0.664	0.663
30	**0.670**	0.665	0.664
35	**0.670**	0.665	0.665
40	**0.670**	0.666	0.665
45	**0.670**	0.666	0.664
50	**0.669**	0.666	0.664

(b) Novelty (NOV)

N	theta_f=0.5	theta_f=0.6	theta_f=0.8
5	**0.983**	0.981	0.981
10	**0.983**	0.983	0.981
15	0.981	**0.983**	0.980
20	0.980	**0.984**	0.981
25	0.980	**0.984**	0.980
30	0.981	**0.984**	0.981
35	0.981	**0.984**	0.980
40	0.979	**0.984**	0.980
45	0.980	**0.984**	0.981
50	0.979	**0.984**	0.981

Seed 2024; single-run values. No across-seed standard deviation is available.

2CKG4ER across forgetting thresholds

theta_m=0.8, rec=50; same seed and model; higher values rank better

(c) Ep, response threshold = 0.4

N	theta_f=0.5	theta_f=0.6	theta_f=0.8
5	**0.036**	-0.027	-0.010
10	**0.072**	-0.072	-0.149
15	**0.025**	-0.105	-0.080
20	-0.056	-0.063	**-0.005**
25	-0.026	-0.092	**0.001**
30	0.032	-0.095	**0.106**
35	0.112	-0.107	**0.128**
40	**0.134**	-0.101	0.126
45	**0.130**	-0.167	0.120
50	**0.123**	-0.174	0.106

(d) Ep, response threshold = 0.5

N	theta_f=0.5	theta_f=0.6	theta_f=0.8
5	**-0.024**	-0.099	-0.097
10	**-0.057**	-0.204	-0.275
15	**-0.174**	-0.318	-0.293
20	-0.282	-0.351	**-0.266**
25	**-0.286**	-0.430	-0.307
30	**-0.244**	-0.501	-0.256
35	**-0.185**	-0.572	-0.274
40	**-0.176**	-0.589	-0.308
45	**-0.175**	-0.662	-0.322
50	**-0.181**	-0.693	-0.343

Seed 2024; single-run values. No across-seed standard deviation is available.

2CKG4ER across forgetting thresholds

theta_m=0.8, rec=50; same seed and model; higher values rank better

(e) Ep, response threshold = 0.6

N	theta_f=0.5	theta_f=0.6	theta_f=0.8
5	**-0.089**	-0.175	-0.166
10	**-0.180**	-0.338	-0.391
15	**-0.343**	-0.491	-0.517
20	**-0.462**	-0.535	-0.540
25	**-0.483**	-0.637	-0.606
30	**-0.503**	-0.714	-0.582
35	**-0.492**	-0.768	-0.645
40	**-0.473**	-0.777	-0.697
45	**-0.503**	-0.809	-0.725
50	**-0.527**	-0.808	-0.746

(f) Ep, response threshold = 0.7

N	theta_f=0.5	theta_f=0.6	theta_f=0.8
5	**-0.159**	-0.258	-0.232
10	**-0.310**	-0.486	-0.540
15	**-0.551**	-0.632	-0.647
20	-0.701	**-0.670**	-0.675
25	-0.746	-0.743	**-0.721**
30	-0.746	-0.772	**-0.690**
35	-0.729	-0.797	**-0.725**
40	**-0.749**	-0.792	-0.750
45	**-0.723**	-0.815	-0.758
50	**-0.737**	-0.811	-0.768

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 1: theta_m=0.8, theta_f=0.5, rec=50

6/6 models completed | completed | ranked across six models

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.672	0.659	**0.693**	0.659	0.666	0.664
10	0.671	0.659	**0.685**	0.659	0.670	0.664
15	0.669	0.659	**0.680**	0.659	0.672	0.663
20	0.670	0.658	**0.676**	0.660	0.671	0.663
25	0.671	0.658	**0.673**	0.660	0.670	0.662
30	0.670	0.659	**0.672**	0.660	0.670	0.662
35	0.670	0.659	**0.670**	0.660	0.669	0.662
40	**0.670**	0.659	0.669	0.660	0.669	0.662
45	**0.670**	0.659	0.668	0.659	0.670	0.661
50	0.669	0.659	0.668	0.659	**0.670**	0.661

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.983	0.969	**0.987**	0.972	0.985	0.971
10	0.983	0.970	**0.983**	0.974	0.981	0.974
15	0.981	0.970	0.980	0.974	**0.983**	0.975
20	0.980	0.972	0.978	0.976	**0.983**	0.975
25	0.980	0.973	0.977	0.977	**0.981**	0.975
30	0.981	0.973	0.976	0.977	**0.981**	0.976
35	0.981	0.973	0.977	0.978	**0.981**	0.975
40	0.979	0.974	0.978	0.978	**0.980**	0.975
45	0.980	0.975	0.979	0.978	**0.981**	0.975
50	0.979	0.975	0.979	0.978	**0.981**	0.975

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 1: theta_m=0.8, theta_f=0.5, rec=50

6/6 models completed | completed | ranked across six models

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.036	0.008	**0.162**	0.046	0.001	0.035
10	0.072	0.072	**0.279**	0.069	-0.071	0.067
15	0.025	0.032	**0.519**	0.087	-0.049	0.074
20	-0.056	0.067	**0.528**	0.105	-0.150	0.076
25	-0.026	0.134	**0.559**	0.105	-0.075	0.089
30	0.032	0.059	**0.568**	0.100	-0.006	0.098
35	0.112	0.158	**0.550**	0.105	0.046	0.097
40	0.134	0.111	**0.588**	0.108	0.064	0.095
45	0.130	0.129	**0.600**	0.114	0.060	0.107
50	0.123	0.133	**0.610**	0.109	0.013	0.100

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.024	-0.064	**0.103**	-0.023	-0.077	-0.031
10	-0.057	-0.038	**0.162**	-0.056	-0.198	-0.044
15	-0.174	-0.119	**0.343**	-0.093	-0.203	-0.082
20	-0.282	-0.101	**0.331**	-0.126	-0.339	-0.112
25	-0.286	-0.047	**0.296**	-0.157	-0.321	-0.138
30	-0.244	-0.118	**0.279**	-0.192	-0.348	-0.154
35	-0.185	-0.068	**0.210**	-0.205	-0.395	-0.168
40	-0.176	-0.112	**0.211**	-0.221	-0.454	-0.190
45	-0.175	-0.062	**0.202**	-0.234	-0.536	-0.190
50	-0.181	-0.052	**0.186**	-0.255	-0.629	-0.204

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 1: theta_m=0.8, theta_f=0.5, rec=50

6/6 models completed | completed | ranked across six models

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.089	-0.123	**0.055**	-0.099	-0.182	-0.087
10	-0.180	-0.177	**0.032**	-0.184	-0.356	-0.154
15	-0.343	-0.321	**0.171**	-0.278	-0.391	-0.242
20	-0.462	-0.367	**0.148**	-0.344	-0.524	-0.308
25	-0.483	-0.320	**0.078**	-0.401	-0.554	-0.356
30	-0.503	-0.402	**0.010**	-0.443	-0.631	-0.389
35	-0.492	-0.337	**-0.099**	-0.472	-0.640	-0.417
40	-0.473	-0.404	**-0.142**	-0.500	-0.675	-0.445
45	-0.503	-0.333	**-0.176**	-0.516	-0.708	-0.454
50	-0.527	-0.337	**-0.197**	-0.536	-0.788	-0.475

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.159	-0.194	**-0.007**	-0.164	-0.277	-0.161
10	-0.310	-0.325	**-0.123**	-0.296	-0.527	-0.296
15	-0.551	-0.545	**-0.041**	-0.408	-0.604	-0.424
20	-0.701	-0.633	**-0.096**	-0.480	-0.735	-0.514
25	-0.746	-0.602	**-0.213**	-0.549	-0.713	-0.574
30	-0.746	-0.715	**-0.377**	-0.604	-0.736	-0.605
35	-0.729	-0.638	**-0.564**	-0.644	-0.703	-0.631
40	-0.749	-0.744	**-0.622**	-0.675	-0.707	-0.654
45	-0.723	**-0.655**	-0.685	-0.691	-0.724	-0.661
50	-0.737	**-0.676**	-0.712	-0.709	-0.795	-0.678

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 2: theta_m=0.8, theta_f=0.6, rec=50

6/6 models completed | completed | ranked across six models

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.664	0.651	**0.693**	0.663	0.667	0.661
10	0.661	0.648	**0.685**	0.662	0.670	0.662
15	0.664	0.648	**0.679**	0.662	0.668	0.662
20	0.665	0.647	**0.676**	0.661	0.667	0.661
25	0.664	0.648	**0.673**	0.661	0.667	0.662
30	0.665	0.648	**0.671**	0.661	0.666	0.662
35	0.665	0.649	**0.670**	0.661	0.666	0.662
40	0.666	0.651	**0.669**	0.661	0.666	0.661
45	0.666	0.652	**0.668**	0.660	0.666	0.662
50	0.666	0.652	**0.667**	0.660	0.666	0.662

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.981	0.984	**0.987**	0.973	0.985	0.979
10	0.983	**0.984**	0.983	0.972	0.983	0.979
15	0.983	0.984	0.980	0.974	**0.984**	0.978
20	**0.984**	0.984	0.979	0.974	0.983	0.978
25	**0.984**	0.983	0.977	0.974	0.983	0.978
30	**0.984**	0.983	0.977	0.974	0.983	0.978
35	**0.984**	0.983	0.977	0.974	0.983	0.978
40	**0.984**	0.982	0.978	0.975	0.983	0.978
45	**0.984**	0.981	0.979	0.975	0.982	0.978
50	**0.984**	0.981	0.979	0.976	0.982	0.978

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 2: theta_m=0.8, theta_f=0.6, rec=50

6/6 models completed | completed | ranked across six models

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.027	-0.004	**0.160**	0.045	-0.008	0.043
10	-0.072	0.025	**0.273**	0.090	0.009	0.078
15	-0.105	0.063	**0.510**	0.101	0.001	0.115
20	-0.063	-0.045	**0.538**	0.117	0.006	0.132
25	-0.092	-0.012	**0.543**	0.129	-0.006	0.155
30	-0.095	0.008	**0.578**	0.129	0.010	0.164
35	-0.107	-0.007	**0.573**	0.126	0.014	0.173
40	-0.101	-0.039	**0.587**	0.122	0.025	0.181
45	-0.167	0.103	**0.607**	0.125	0.014	0.176
50	-0.174	0.137	**0.605**	0.123	-0.044	0.178

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.099	-0.082	**0.110**	-0.027	-0.075	-0.024
10	-0.204	-0.101	**0.156**	-0.037	-0.138	-0.044
15	-0.318	-0.105	**0.331**	-0.072	-0.252	-0.062
20	-0.351	-0.243	**0.333**	-0.088	-0.327	-0.080
25	-0.430	-0.234	**0.285**	-0.113	-0.398	-0.080
30	-0.501	-0.250	**0.277**	-0.132	-0.441	-0.097
35	-0.572	-0.245	**0.235**	-0.152	-0.478	-0.111
40	-0.589	-0.257	**0.236**	-0.170	-0.502	-0.117
45	-0.662	-0.166	**0.218**	-0.184	-0.524	-0.128
50	-0.693	-0.152	**0.183**	-0.198	-0.575	-0.140

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 2: theta_m=0.8, theta_f=0.6, rec=50

6/6 models completed | completed | ranked across six models

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.175	-0.160	**0.061**	-0.100	-0.150	-0.091
10	-0.338	-0.221	**0.029**	-0.161	-0.276	-0.171
15	-0.491	-0.272	**0.156**	-0.236	-0.431	-0.221
20	-0.535	-0.454	**0.139**	-0.294	-0.549	-0.275
25	-0.637	-0.496	**0.053**	-0.345	-0.628	-0.314
30	-0.714	-0.543	**0.006**	-0.386	-0.669	-0.349
35	-0.768	-0.583	**-0.076**	-0.423	-0.708	-0.373
40	-0.777	-0.611	**-0.106**	-0.450	-0.721	-0.397
45	-0.809	-0.585	**-0.125**	-0.471	-0.739	-0.421
50	-0.808	-0.576	**-0.174**	-0.491	-0.784	-0.440

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.258	-0.242	**-0.004**	-0.164	-0.195	-0.154
10	-0.486	-0.346	**-0.125**	-0.282	-0.371	-0.297
15	-0.632	-0.414	**-0.049**	-0.393	-0.554	-0.410
20	-0.670	-0.620	**-0.122**	-0.475	-0.658	-0.497
25	-0.743	-0.683	**-0.213**	-0.537	-0.699	-0.551
30	-0.772	-0.728	**-0.334**	-0.580	-0.718	-0.598
35	-0.797	-0.743	**-0.476**	-0.610	-0.739	-0.624
40	-0.792	-0.740	**-0.547**	-0.646	-0.740	-0.641
45	-0.815	-0.701	**-0.592**	-0.671	-0.748	-0.662
50	-0.811	-0.697	**-0.663**	-0.692	-0.788	-0.676

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 3: theta_m=0.8, theta_f=0.8, rec=50

6/6 models completed | completed | ranked across six models

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.664	0.655	**0.693**	0.667	0.663	0.664
10	0.661	0.654	**0.685**	0.665	0.664	0.663
15	0.662	0.654	**0.680**	0.664	0.664	0.662
20	0.663	0.655	**0.676**	0.663	0.665	0.662
25	0.663	0.656	**0.674**	0.662	0.664	0.662
30	0.664	0.656	**0.672**	0.662	0.665	0.663
35	0.665	0.656	**0.670**	0.661	0.665	0.662
40	0.665	0.656	**0.669**	0.660	0.665	0.662
45	0.664	0.656	**0.669**	0.660	0.664	0.662
50	0.664	0.655	**0.668**	0.659	0.664	0.662

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.981	0.986	**0.987**	0.987	0.982	0.975
10	0.981	**0.987**	0.983	0.985	0.983	0.977
15	0.980	**0.986**	0.980	0.985	0.981	0.977
20	0.981	**0.985**	0.978	0.985	0.981	0.978
25	0.980	**0.985**	0.977	0.985	0.982	0.978
30	0.981	**0.985**	0.976	0.985	0.982	0.980
35	0.980	**0.985**	0.977	0.985	0.981	0.980
40	0.980	**0.985**	0.978	0.985	0.981	0.980
45	0.981	**0.985**	0.979	0.984	0.981	0.981
50	0.981	**0.984**	0.980	0.984	0.981	0.980

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 3: theta_m=0.8, theta_f=0.8, rec=50

6/6 models completed | completed | ranked across six models

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.010	**0.270**	0.159	0.044	-0.060	0.038
10	-0.149	**0.422**	0.275	0.079	-0.078	0.065
15	-0.080	0.487	**0.498**	0.090	-0.160	0.101
20	-0.005	0.507	**0.530**	0.090	-0.148	0.115
25	0.001	0.489	**0.546**	0.103	-0.134	0.128
30	0.106	0.454	**0.587**	0.097	-0.042	0.137
35	0.128	0.436	**0.555**	0.101	-0.030	0.146
40	0.126	0.462	**0.606**	0.102	-0.026	0.151
45	0.120	0.403	**0.591**	0.107	-0.028	0.158
50	0.106	0.401	**0.581**	0.114	0.024	0.163

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.097	**0.195**	0.110	-0.024	-0.130	-0.039
10	-0.275	**0.295**	0.156	-0.054	-0.216	-0.058
15	-0.293	**0.337**	0.323	-0.087	-0.353	-0.072
20	-0.266	0.310	**0.322**	-0.123	-0.380	-0.098
25	-0.307	0.244	**0.289**	-0.158	-0.418	-0.119
30	-0.256	0.176	**0.310**	-0.189	-0.386	-0.133
35	-0.274	0.123	**0.253**	-0.210	-0.414	-0.149
40	-0.308	0.115	**0.280**	-0.226	-0.437	-0.160
45	-0.322	0.031	**0.219**	-0.236	-0.455	-0.165
50	-0.343	0.022	**0.174**	-0.253	-0.409	-0.182

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 3: theta_m=0.8, theta_f=0.8, rec=50

6/6 models completed | completed | ranked across six models

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.166	**0.136**	0.059	-0.098	-0.191	-0.107
10	-0.391	**0.202**	0.026	-0.181	-0.311	-0.181
15	-0.517	**0.204**	0.143	-0.254	-0.491	-0.234
20	-0.540	0.125	**0.137**	-0.323	-0.540	-0.290
25	-0.606	0.019	**0.052**	-0.371	-0.602	-0.335
30	-0.582	-0.071	**0.020**	-0.420	-0.581	-0.365
35	-0.645	-0.176	**-0.072**	-0.455	-0.609	-0.389
40	-0.697	-0.251	**-0.091**	-0.496	-0.629	-0.424
45	-0.725	-0.395	**-0.152**	-0.523	-0.641	-0.440
50	-0.746	-0.468	**-0.195**	-0.543	-0.597	-0.458

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.232	**0.056**	-0.003	-0.175	-0.251	-0.165
10	-0.540	**0.084**	-0.124	-0.304	-0.406	-0.300
15	-0.647	**0.035**	-0.062	-0.404	-0.589	-0.389
20	-0.675	**-0.098**	-0.113	-0.485	-0.649	-0.477
25	-0.721	**-0.216**	-0.230	-0.539	-0.712	-0.533
30	-0.690	-0.364	**-0.328**	-0.591	-0.692	-0.570
35	-0.725	-0.504	**-0.487**	-0.631	-0.718	-0.594
40	-0.750	-0.608	**-0.539**	-0.661	-0.735	-0.622
45	-0.758	-0.717	-0.643	-0.675	-0.740	**-0.636**
50	-0.768	-0.752	-0.707	-0.684	-0.696	**-0.648**

Seed 2024; single-run values. No across-seed standard deviation is available.